Chapter 1: Materials and Bonding

1.1 Introduction

Materials Science is an interdisciplinary field concerned with understanding how the composition, structure, processing, and properties of a material are related to one another and how these relationships determine its performance in practical applications. In simple terms, Materials Science seeks to answer a fundamental question: Why does a material behave the way it does?

The structure of a material refers to the arrangement and organization of its constituent elements at different length scales, ranging from the electronic and atomic levels to the microstructural and macroscopic levels. Changes in composition or structure can therefore produce significant changes in the physical and chemical behavior of a material. For example, differences in crystal structure, grain size, defects, or porosity can strongly influence its mechanical, electrical, thermal, magnetic, and optical properties.

Processing is also an essential component of Materials Science because the method used to synthesize or manufacture a material determines, to a large extent, its final structure. Consequently, the properties of a material are not determined only by its chemical composition; they also depend on how its structure is produced and modified during processing.

This interrelationship among *processing*, *structure*, *properties*, and *performance* of materials may be depicted in linear fashion as in the schematic illustration:



Figure 1.1. The four components of the discipline of materials science and engineering and their interrelationship.

The main classes of material properties include mechanical, electrical, thermal, magnetic, optical, and chemical properties. These properties describe how a material responds to different external stimuli or environmental conditions. The combination of these properties ultimately determines the performance of a material in a particular application.

Understanding these relationships provides the scientific basis for selecting existing materials, modifying their properties, and developing new materials for specific technological applications.

An Example: Processing, Structure, and Properties

A simple example illustrates the relationship between processing, structure, and properties. The three aluminum oxide (Al_2O_3) specimens shown in Figure 1.2 have the same chemical composition but exhibit different optical properties: one is transparent, another translucent, and the third opaque. These differences arise from variations in their internal structure, particularly the presence of grain

boundaries and pores. The microstructure, in turn, is strongly influenced by the processing conditions used to produce the material.

Thus, even when the chemical composition is essentially the same, changes in processing can produce different structures and, consequently, different properties.

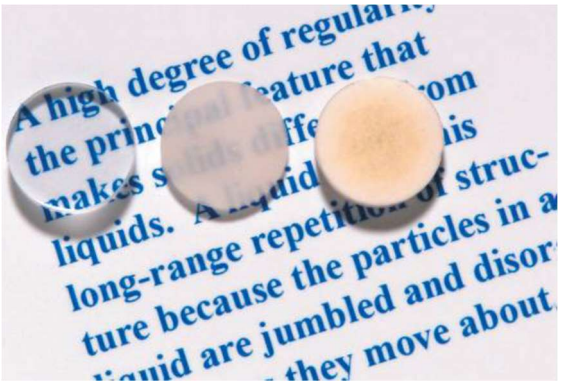


Figure 1.2. Aluminum oxide (Al_2O_3) specimens exhibiting different optical properties: transparent (left), translucent (center), and opaque (right). The differences in light transmission result from variations in microstructure produced by different processing conditions.

1.2 Classification of Materials

Materials can be classified according to their chemical composition, atomic bonding, structure, processing, and functional properties. Although no single classification is sufficient for all purposes, the principal classes of engineering and functional materials are metals and alloys, ceramics and glasses, polymers, semiconductors, and composites. In addition, materials may be classified according to their applications or characteristic structural scale, giving rise to important groups such as biomaterials, smart materials, and nanomaterials.

1.2.1 Metals and Alloys

Metals are predominantly metallic elements bonded mainly through metallic bonding, while ***alloys*** contain two or more elements, at least one of which is metallic.

Metallic materials have large numbers of nonlocalized electrons—that is, these electrons are not bound to particular atoms. Many properties of metals are directly attributable to these electrons. For example, metals are extremely good conductors of electricity and heat, and are not transparent to visible light; a polished metal surface has a lustrous appearance. In addition, some of the metals (i.e., Fe, Co, and Ni) have desirable magnetic properties.

Examples: iron (Fe), aluminum (Al), copper (Cu), titanium (Ti), nickel (Ni), and alloys such as steel, stainless steel, and brass.

Typical applications: structural components, electrical conductors, transportation systems, and machinery.

1.2.2 Ceramics and Glasses

Ceramics are compounds between metallic and nonmetallic elements; they are most frequently oxides, nitrides, and carbides. Some refer to as the traditional ceramics—those composed of clay minerals as well as cement and glass.

Ceramic materials commonly contain strong ionic and/or covalent bonds. Many ceramics exhibit high hardness, stiffness, wear resistance, and thermal stability, but conventional ceramics are often brittle because of their limited ability to accommodate plastic deformation (lack of ductility).

Ceramic materials are typically insulator of heat and electricity (i.e., have low electrical conductivities) and are more resistant to high temperatures. With regard to optical characteristics, ceramics may be transparent, translucent, or opaque, and some of the oxide ceramics (e.g., Fe_3O_4) exhibit magnetic behavior.

Examples: alumina (Al_2O_3), zirconia (ZrO_2), silica (SiO_2), silicon carbide (SiC), silicon nitride (Si_3N_4), glass, and cement-based materials.

Typical applications: refractories, cutting tools, electrical insulators, and protective coatings.

1.2.3 Polymers

Polymers are macromolecular materials composed of long chains or networks formed from repeating molecular units. Polymers include the familiar plastic and rubber materials. Many of them are organic compounds that are chemically based on carbon, hydrogen, and other nonmetallic elements (i.e., O, N, and Si).

Their bonding and molecular architecture give them generally low density, relatively low stiffness, and low thermal and electrical conductivity compared with most metals. Many polymers are highly formable and ductile, while their mechanical and thermal behavior can be strongly influenced by molecular structure, crystallinity, cross-linking, and temperature.

Examples: polyethylene (PE), polypropylene (PP), poly(vinyl chloride) (PVC), polystyrene (PS), nylon, epoxy, and silicone rubber.

Typical applications: packaging, electrical insulation, structural components, coatings, fibers, and biomedical devices.

1.2.4 Composites

Composites are materials formed by combining two or more individual materials (metals, ceramics, and polymers) to obtain a tailored combination of properties. A composite commonly consists of a *matrix* that surrounds and supports a *reinforcement* phase. The properties of the resulting material depend not only on the constituents but also on their volume fractions, morphology, orientation, and interfacial bonding.

One of the most common and familiar composites is glass–fiber–reinforced polymer (GFRP), in which small glass fibers are embedded within a polymeric material (normally an epoxy or polyester). The glass fibers are relatively strong and stiff (but also brittle), whereas the polymer is more flexible. Thus, GFRP is relatively stiff, strong, and flexible. In addition, it has a low density.

Another technologically important material is the carbon–fiber–reinforced polymer (CFRP) composite, carbon fibers that are embedded within a polymer. These materials are stiffer and stronger than glass fiber–reinforced materials but more expensive. CFRP composites are used in some aircraft and aerospace applications, as well as in high-tech sporting equipment.

Examples: glass-fiber-reinforced polymer (GFRP), carbon-fiber-reinforced polymer (CFRP), concrete, and ceramic-matrix composites.

Typical applications: aerospace structures, automobiles, sporting equipment, construction, and protective components.

The following figures illustrate and compare selected properties of these material classes:

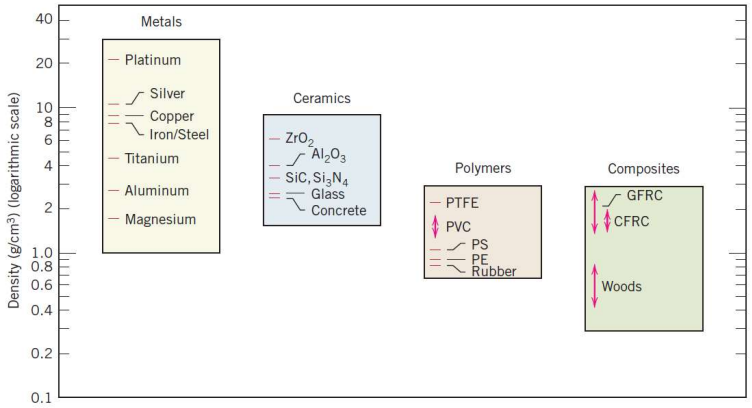


Figure 1.3. Bar chart of room temperature Density values for various metals, ceramics, polymers, and composite materials.

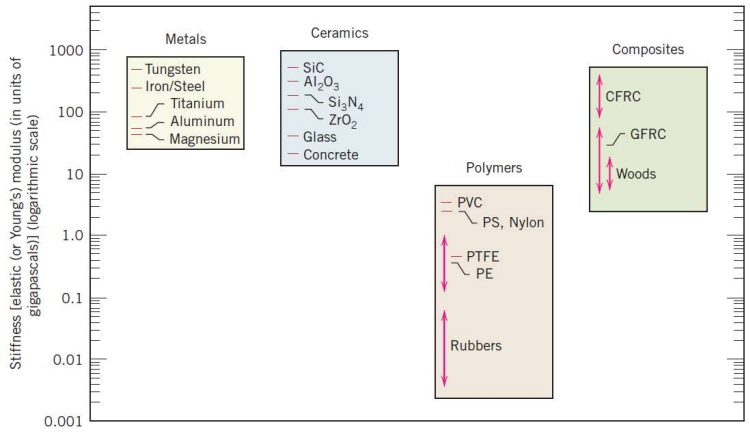


Figure 1.4. Bar chart of room temperature Stiffness (i.e., elastic modulus) values for various metals, ceramics, polymers, and composite materials.

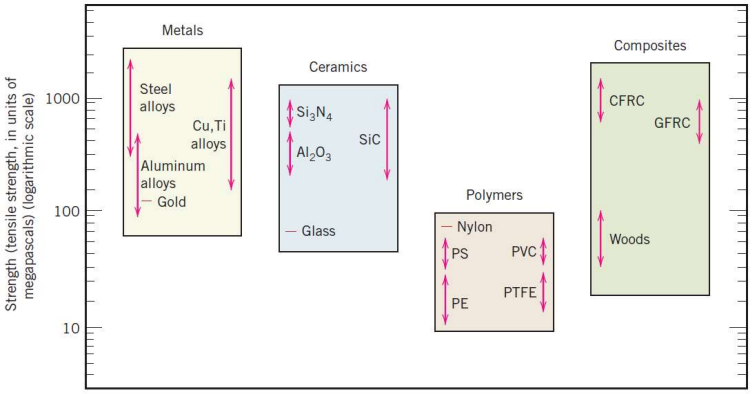


Figure 1.5. Bar chart of room temperature strength (i.e., tensile strength) values for various metals, ceramics, polymers, and composite materials.

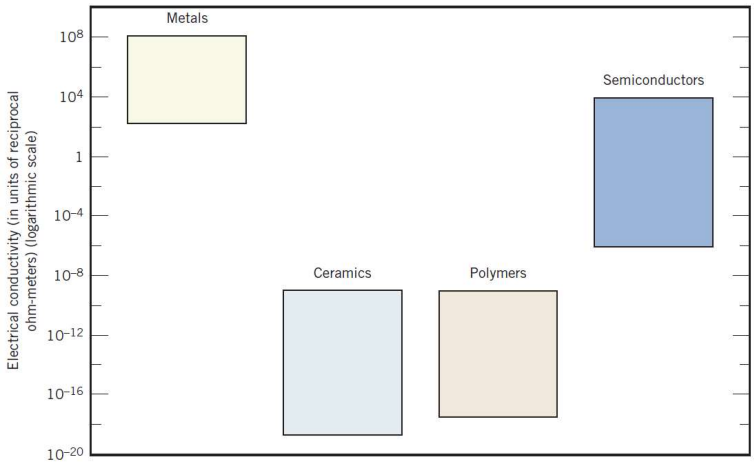


Figure 1.6. Bar chart of room temperature Electrical conductivity ranges for metals, ceramics, polymers, and semiconducting materials.

1.3 Advanced Materials

Advanced materials are materials developed or engineered to provide enhanced or specialized properties for high-technology applications. They may be metals, ceramics, polymers, composites,

or other material systems, and their performance is often achieved through controlled composition, structure, processing, or nanoscale design.

The main categories considered in this section are:

Semiconductors, Biomaterials, Smart and Functional Materials, and Nanomaterials.

1.3.1 Semiconductors

Semiconductors have electrical properties that are intermediate between those of electrical conductors (i.e., metals and metal alloys) and insulators (i.e., ceramics and polymers). The electrical characteristics of these materials are extremely sensitive to the presence of minute concentrations of impurity atoms, for which the concentrations may be controlled over very small spatial regions.

Examples: silicon (Si), germanium (Ge), gallium arsenide (GaAs), gallium nitride (GaN), and silicon carbide (SiC).

Typical applications: transistors, integrated circuits, solar cells, light-emitting diodes, lasers, and power electronics.

1.3.2 Biomaterials

Biomaterials are materials designed to interact with biological systems for a medical or biomedical purpose. Their critical requirement is appropriate *biocompatibility*, together with the mechanical, chemical, biological, and surface properties required for a specific application. Biomaterials are not a separate chemical class; they may be metals, ceramics, polymers, or composites.

Examples: titanium alloys for orthopedic implants, alumina and zirconia for biomedical components, and polyethylene for joint replacements.

Typical applications: orthopedic implants, dental materials, tissue engineering, and drug-delivery systems.

1.3.3 Smart Materials

Smart materials are a group of new materials capable of responding to changes in their environment through a predictable and useful change in one or more properties. These materials are able to sense changes in their environment and then respond to these changes in predetermined manners. Their response may be triggered by temperature, electric fields, magnetic fields, mechanical stress, or light.

Four types of materials are commonly used for actuators:

1. *Shape-memory alloys* are metals that, after having been deformed, revert to their original shape when temperature is changed.
2. *Piezoelectric ceramics* expand and contract in response to an applied electric field (or voltage); conversely, they also generate an electric field when their dimensions are altered.
3. *Magnetostrictive materials* is analogous to that of the piezoelectrics, except that they are responsive to magnetic fields.
4. *Electrorheological and magnetorheological fluids* are liquids that experience dramatic changes in viscosity upon the application of electric and magnetic fields, respectively.

Examples: shape-memory alloys, piezoelectric ceramics, magnetostrictive materials, and magnetorheological or electrorheological materials.

Typical applications: sensors, actuators, adaptive structures, energy harvesting, and electronic or optical devices.

1.3.4 Nanomaterials

Nanomaterials are materials having at least one characteristic structural dimension in the nanometer scale (*approximately 1–100 nm*). At this scale, materials can exhibit properties that differ significantly from their bulk counterparts because of size-dependent effects, including a high surface-to-volume ratio and, in some systems, quantum effects.

Nanomaterials are not a separate chemical class; metals, ceramics, polymers, semiconductors, and composites can all be engineered at the nanoscale. Nanoparticles can generally be prepared by two main approaches:

Top-down approach: A bulk material is progressively broken down or reduced in size to produce nanoscale structures.

Bottom-up approach: Nanostructures are built from atoms, ions, molecules, or other nanoscale building blocks through controlled chemical or physical processes.

Examples: nanoparticles, quantum dots, nanowires, carbon nanotubes, graphene, and nanocomposites.

Typical applications: electronics, energy storage and conversion, sensors, catalysis, coatings, medicine, and advanced coatings.

1.4 Atomic Structure

An atom consists of a small, positively charged nucleus, containing protons and neutrons, surrounded by electrons. Electrons occupy discrete energy levels (states) and are distributed among shells and subshells. The electrons in the outermost occupied shell are called valence electrons. They are particularly important in Materials Science because they participate in interactions between

atoms and determine the nature of atomic bonding, as well as many of the physical and chemical properties of materials.

1.5 Bonding Forces and Energies

When two atoms approach each other, they experience *attractive* and *repulsive* interatomic forces. At large separation distances, these interactions are negligible. As the atoms approach, attractive forces associated with atomic bonding become significant, while strong repulsive forces arise when their electron clouds begin to overlap.

The net force (F_N) between two atoms is the sum of the attractive (F_A) and repulsive (F_R) forces:

$$F_N = F_A + F_R$$

At the equilibrium interatomic distance r_o, the attractive and repulsive forces are equal in magnitude and opposite in direction, so the net force is zero. At this distance, the atoms are in a stable equilibrium state. For many atoms, r_o is approximately 0.3 nm.

The interaction can also be described in terms of **potential energy**. The minimum of the potential-energy curve corresponds to the equilibrium distance r_o, and the depth of this minimum represents the **bonding energy** E_o (the energy required to separate the bonded atoms).

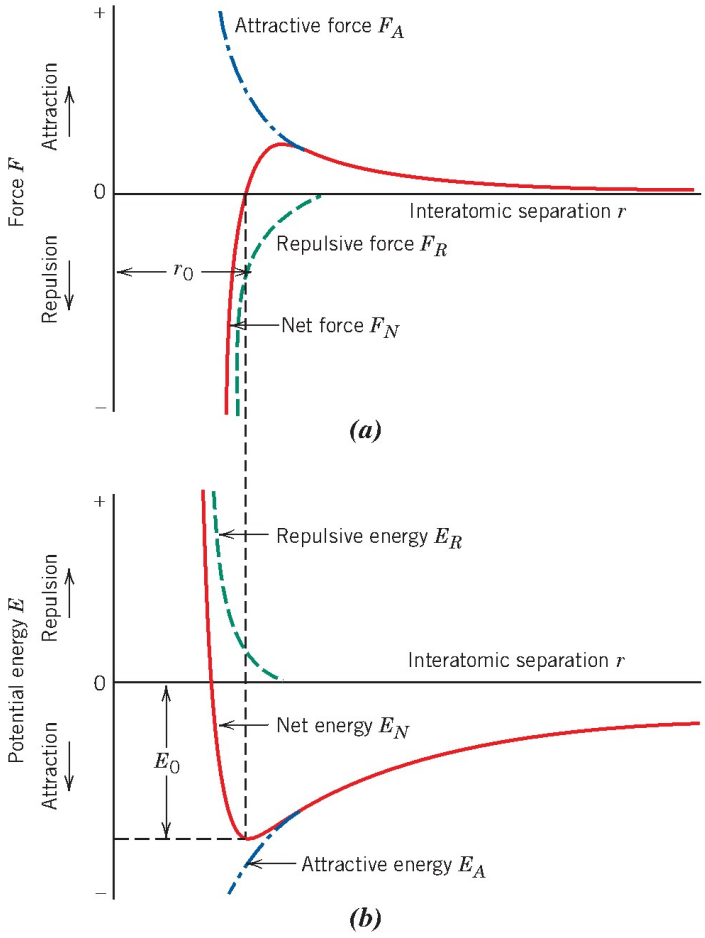


Figure 1.7.

(a) The dependence of repulsive, attractive, and net forces on interatomic separation for two isolated atoms.

(b) The dependence of repulsive, attractive, and net potential energies on interatomic separation for two isolated atoms.

The magnitude of the bonding energy and the shape of the energy–distance curve influence important material properties. In general, stronger atomic bonding is associated with higher melting temperatures and greater stiffness.

Atomic bonding in solids is commonly divided into **primary (chemical) bonds** (ionic, covalent, and metallic) and **secondary (physical) bonds**, such as van der Waals interactions. Primary bonding involves the valence electrons of the constituent atoms.

1.6 Primary Interatomic Bonds

1.6.1 Ionic Bonding

Ionic bonding results from the transfer of one or more valence electrons from one atom to another, typically between atoms with substantially different electronegativities. The atom that loses electrons becomes a positively charged **cation**, while the atom that gains electrons becomes a negatively charged **anion**. This electron transfer often leads to more stable electron configurations.

A classic example is sodium chloride (**NaCl**). A sodium atom transfers its valence electron to a chlorine atom, forming (Na^+) and (Cl^-) ions. The oppositely charged ions are then held together by *Coulombic electrostatic attraction*.

Ionic bonding is **nondirectional**, meaning that the electrostatic attraction acts in all directions around an ion. Consequently, ionic solids tend to form crystal structures in which each ion is surrounded by oppositely charged ions.

Ionic bonds are generally strong, and many ionic solids therefore have high melting temperatures and high hardness. They are also typically brittle and have low electrical and thermal conductivity in the solid state because their ions are not freely mobile.

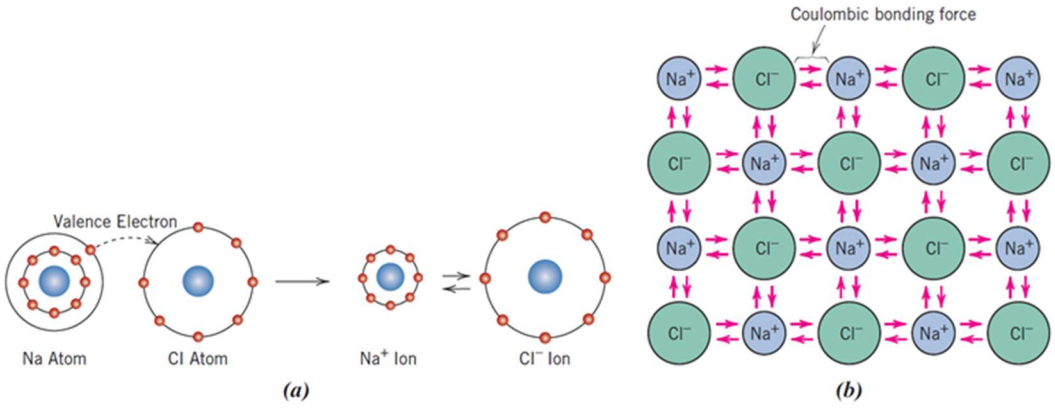


Figure 1.8. Schematic representations of **(a)** the formation of Na^+ and Cl^- ions and **(b)** ionic bonding in sodium chloride (NaCl).

1.6.2 Covalent Bonding

Covalent bonding occurs when atoms share one or more pairs of valence electrons to achieve a more stable electron configuration. Each participating atom contributes electrons to the bond, and the shared electrons are attracted to the nuclei of both atoms. The bond is associated with overlap of atomic orbitals between the participating atoms.

Unlike ionic bonding, covalent bonding is generally **directional**. The shared electrons are concentrated in specific regions between particular atoms, so the direction and geometry of the bonds strongly influence the structure and properties of covalently bonded materials.

Covalent bonding occurs in many molecules, such as (H_2), (Cl_2), and (H_2O), and in important solid materials such as diamond, silicon, germanium, GaAs, and SiC.

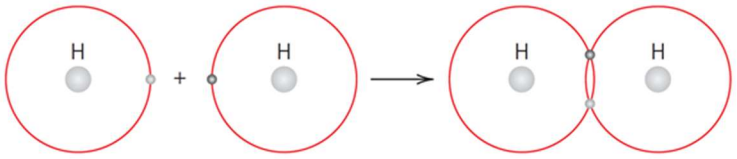


Figure 1.9. Schematic representation of covalent bonding in a molecule of hydrogen (H_2).

Covalent bonds can be strong, as in diamond, and their properties depend strongly on the atomic arrangement and bonding environment. Because the electrons involved in covalent bonds are relatively localized, many covalently bonded materials are electrical insulators or semiconductors. Their mechanical properties vary widely depending on their structure and bonding.

Bond Hybridization in Carbon

Hybridization is the mixing of atomic orbitals to form new hybrid orbitals that are oriented to produce effective covalent bonding. Carbon provides an important example.

In **sp^3 hybridization**, one (s) orbital combines with three (p) orbitals to form four equivalent sp^3 hybrid orbitals. These orbitals are directed toward the corners of a tetrahedron, with bond angles of approximately 109.5°. This arrangement occurs in diamond, where each carbon atom forms four strong covalent bonds with neighboring carbon atoms. The resulting three-dimensional network contributes to diamond's exceptional hardness and high thermal stability.

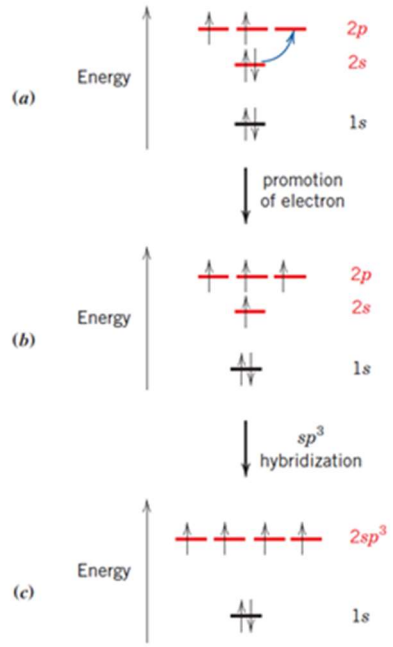


Figure 1.10. Schematic diagram that shows the formation of sp^3 hybrid orbitals in carbon.
(a) Promotion of a 2s electron to a 2p state;
(b) this promoted electron in a 2p state;
(c) four $2sp^3$ orbitals that form by mixing the single 2s orbital with the three 2p orbitals.

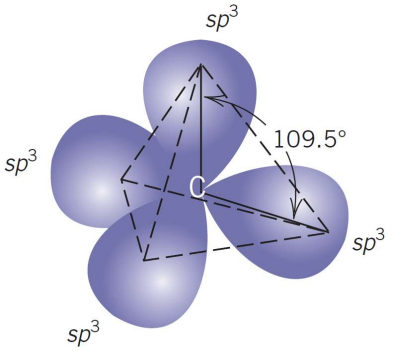


Figure 1.11. Schematic diagram showing four sp3 hybrid orbitals that point to the corners of a tetrahedron; the angle between orbitals is 109.5°.

In **sp² hybridization**, one (s) orbital combines with two (p) orbitals to form three sp² hybrid orbitals arranged in a plane with bond angles of approximately 120°. One (p) orbital remains unhybridized and is oriented perpendicular to this plane. This bonding arrangement occurs in graphite, where carbon atoms form strong covalent bonds within two-dimensional layers, while the layers are held together by much weaker van der Waals interactions.

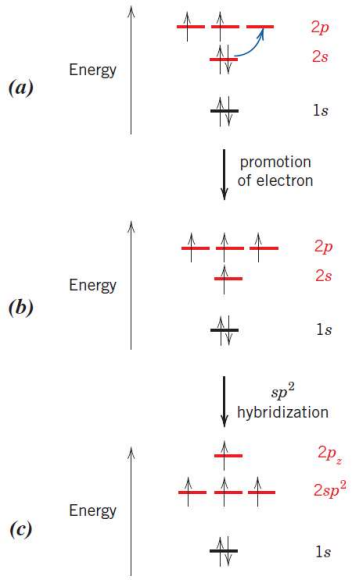


Figure 1.12. Schematic diagram that shows the formation of sp2 hybrid orbitals in carbon.

(a) Pro motion of a 2s electron to a 2p state;

(b) this promoted electron in a 2p state;

(c) three $2sp^2$ orbitals that form by mixing the single 2s orbital with two 2p orbitals—the $2p_z$ orbital remains unhybridized.

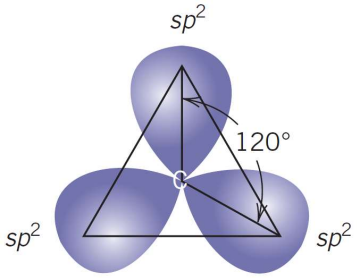


Figure 1.13. Schematic diagram showing three sp^2 orbitals that are coplanar and point to the corners of a triangle; the angle between adjacent orbitals is 120°.

Thus, **diamond** and **graphite** illustrate an important Materials Science principle: differences in atomic bonding and structure can produce very different material properties even when the materials have the same chemical composition.

1.6.3 Metallic Bonding

Metallic bonding is characteristic of metals and their alloys. In a simple model, the valence electrons are delocalized and are not associated with any particular atom. Instead, they move relatively freely throughout the solid, forming an "*electron sea*" surrounding positively charged ion cores. Each ion core consists of the atomic nucleus and the inner, nonvalence electrons.

The electrostatic attraction between the positively charged ion cores and the delocalized electrons provides the cohesive force that holds the metal together. Because the delocalized electrons can move throughout the solid, metallic bonding is **nondirectional**. This allows the atoms to change their relative positions without completely disrupting the bonding, which contributes to the characteristic ductility of metals.

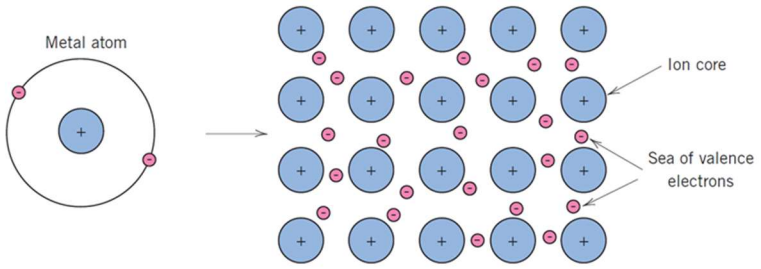


Figure 1.14. Schematic illustration of metallic bonding.

The presence of mobile electrons also explains two important properties of metals:

High electrical conductivity, because electrons can move through the material under an applied electric field.

High thermal conductivity, because mobile electrons can efficiently transport energy through the solid.

Comparison

Bond type	Electrons	Directionality	Important consequence
Ionic	Transferred	Nondirectional	Strong, often brittle
Covalent	Shared/localized	Directional	Strong, often stiff/brittle
Metallic	Delocalized	Nondirectional	Electrical/thermal conductivity and ductility

1.7 Secondary Bonds

Secondary bonds, also called physical or intermolecular bonds, are weaker than primary ionic, covalent, and metallic bonds. They arise mainly from electrostatic interactions between permanent or temporary electric dipoles and are present between atoms or molecules when primary bonding does not dominate the interaction.

An *electric dipole* occurs when positive and negative charge distributions are separated within an atom or molecule. Secondary interactions can therefore occur between temporary dipoles, between temporary and permanent dipoles, or between permanent dipoles.

1.7.1 Van der Waals Bonding

Van der Waals Bonding arise from interactions between temporary and/or permanent molecular dipoles. They are generally weaker than primary bonds and are important in many molecular and layered materials.

Three common types can be distinguished:

London dispersion interactions: Random fluctuations in the electron distribution of an atom or molecule produce an instantaneous dipole, which can induce a dipole in a neighboring atom or

molecule. These interactions occur even between nonpolar atoms and molecules and are responsible, for example, for the condensation of noble gases.

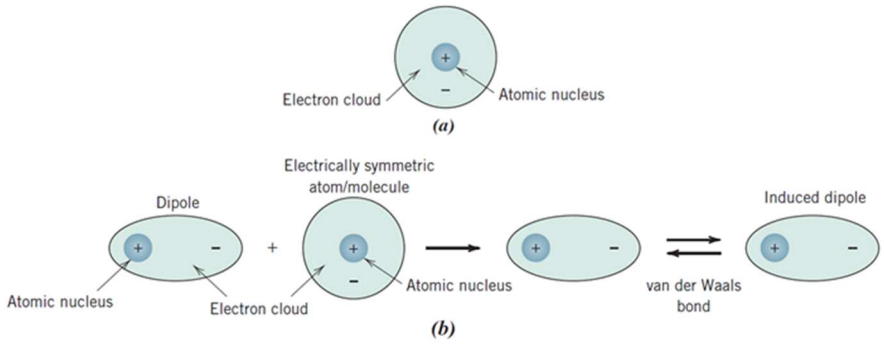


Figure 1.15. Schematic representations of **(a)** an electrically symmetric atom and **(b)** how an electric dipole induces an electrically symmetric atom/molecule to become a dipole, and the van der Waals bond between the dipoles.

Dipole–induced dipole interactions: A molecule with a permanent dipole can distort the electron distribution of a neighboring nonpolar atom or molecule, inducing a dipole. The resulting attraction is generally stronger than a purely instantaneous dipole interaction.

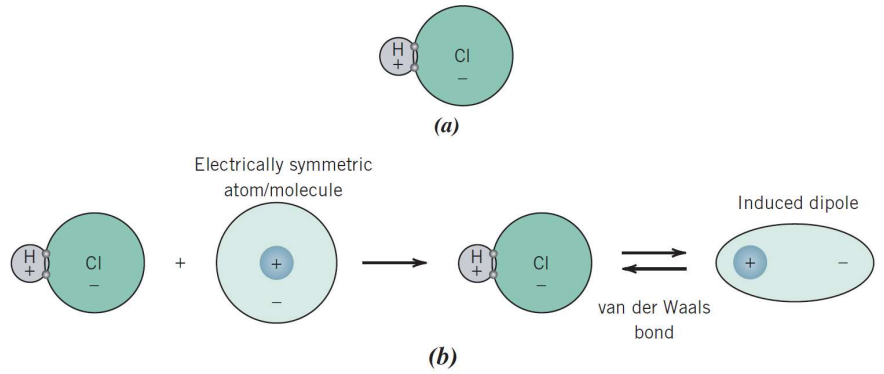


Figure 1.16. Schematic representations of **(a)** a hydrogen chloride molecule (dipole) and **(b)** how an HCl molecule induces an electrically symmetric atom/ molecule to become a dipole, and the van der Waals bond between these dipoles.

Dipole–dipole interactions: Permanent dipoles in polar molecules can interact through electrostatic attraction between oppositely charged ends of neighboring molecules.

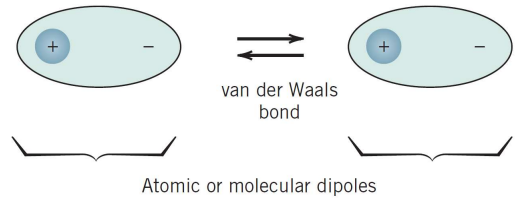


Figure 1.17. Schematic illustration of van der Waals bonding between two dipoles.

Van der Waals interactions are generally **nondirectional** or relatively weakly directional compared with covalent bonds, and their cumulative effect can become significant when large numbers of atoms or molecules interact.

1.7.2 Hydrogen Bonding

Hydrogen bonding is a relatively strong and more directional type of secondary interaction. It occurs when hydrogen is covalently bonded to a highly electronegative atom, particularly fluorine, oxygen, or nitrogen, and interacts with an electronegative atom in a neighboring molecule or molecular group.

For example, hydrogen bonding occurs between water molecules. The strong polarity of the O–H bond produces a partially positive hydrogen atom and a partially negative oxygen atom. The hydrogen atom can then interact attractively with an oxygen atom of a neighboring molecule.

Hydrogen bonding can significantly affect melting and boiling temperatures, viscosity, solubility, and the mechanical behavior of polymeric materials.

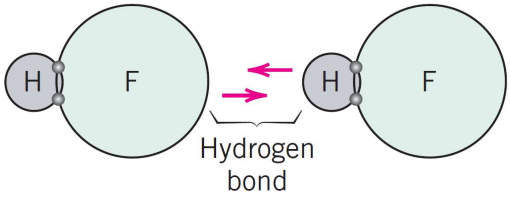


Figure 1.18. Schematic representation of hydrogen bonding in hydrogen fluoride (HF).

Although secondary bonds are much weaker than primary bonds, their collective effect can strongly influence the structure and properties of materials. Thus, material properties are determined not only by the type of primary bonding but also by the secondary interactions between atoms, molecules, or structural units.

1.8 Mixed Bonding

The four bonding types introduced above (ionic, covalent, metallic, and secondary bonding) represent idealized descriptions of interatomic interactions. In many real materials, however, bonding cannot be described by a single pure type. Instead, a bond may exhibit a combination of two or more types, referred to as ***mixed bonding***.

The most common mixed character occurs between ionic and covalent bonding. Most covalent bonds have some degree of ionic character, and many ionic bonds have some covalent character. The relative contribution of each depends largely on the difference in electronegativity between the constituent atoms. A larger electronegativity difference generally corresponds to greater ionic character, whereas a smaller difference is associated with greater covalent character.

Mixed covalent–metallic bonding also occurs in materials whose properties lie between those of typical metals and nonmetals.

The type and degree of bonding strongly influence the material's crystal structure, electronic behavior, mechanical properties, melting temperature, and other physical properties.

1.9 Molecules and Molecular Materials

A ***molecule*** is a group of two or more atoms held together by strong primary bonds, most commonly covalent bonds. Examples include elemental diatomic molecules such as (H_2), (O_2), and (F_2), as well as molecular compounds such as (H_2O), (CO_2), (CH_4), and (C_6H_6).

Within a molecule, the atoms are held together by strong intramolecular bonds, whereas interactions between neighboring molecules are generally much weaker secondary interactions, such as van der Waals interactions and hydrogen bonding.

Because the interactions between molecules are relatively weak, many molecular substances have comparatively low melting and boiling temperatures. Small molecular substances are therefore often gases or liquids at ordinary temperatures, although their physical state depends on both molecular structure and the strength of intermolecular interactions.

Molecular materials can also form solids when intermolecular interactions are sufficiently strong or when the molecules are very large. ***Polymers*** are an important example: they consist of very long molecular chains, and their mechanical and thermal properties are influenced by interactions between the chains, including van der Waals interactions and hydrogen bonding.

Thus, molecular materials illustrate an important Materials Science principle: the properties of a material depend not only on the bonds within molecules but also on the interactions between molecules.